

Akindu Delgahagoda

Undergraduate Student

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PERSONAL STATEMENT

Engineering undergraduate with a specialized focus on robotics and distributed systems architecture. Proficient in bridging the gap between embedded hardware and cloud services to create cohesive intelligent solutions. Brings a rigorous, analytical approach to system design, combining theoretical depth with the practical expertise needed to deliver robust, full-stack engineering projects.

EDUCATION

University of Moratuwa

BSc Eng Hons, Computer Science and Engineering
Stream - Integrated Computer Engineering | CGPA - 3.5

Bandaranayake Mawatha, Moratuwa 10400

2022 - 2026

Sabaragamuwa University of Sri Lanka

Diploma in English | Credit Pass

Belihuloya, Ratnapura

2021 - 2022

Sivali Central College

GCE A/L 2020 | Physical Science Stream | 3A Passes
GCE O/L 2017 | 9A Passes

Hidellana, Ratnapura

2007 - 2020

PROFESSIONAL EXPERIENCE

Department of CSE

Teaching Assistant, Robotics and Automation

University of Moratuwa, Katubedds

Jan 2026 - Jun 2026

- Collaborated in creating course material covering ROS2 Theory and practical applications.
- Collaborated in creating quizzes to assess students' understanding of ROS2 concepts.
- Co-developed a new lab series covering fundamental concepts of ROS2 in a step-by-step manner.
- Guided students in developing a robotics project using ROS2, providing mentorship and technical support throughout the development process.

WSO2

Intern Software Engineer, Digital Transformation Team

105, Bauddhaloka Mawatha, Colombo 4

Jan 2025 - Jun 2025

- Developed "Infra Portal," a full-stack infrastructure automation platform using **Ballerina**, **React** and **MySQL** to replace legacy manual workflows, streamlining GitHub repository provisioning across the company.
- Upgraded and optimized existing internal middleware services to facilitate seamless integration with the GitHub API using **GraphQL**, ensuring robust communication between the new portal and external providers.
- Architected the solution to be extensible beyond GitHub, designing a scalable foundation to support future infrastructure automation requirements.
- Implemented secure authentication using **Asgardeo** and deployed the production-ready containerized application on **Choreo** for internal organizational use.

SKILLS

Programming Languages: Python | C/C++ | C# | VHDL | Java | JavaScript | Ballerina

Database: MySQL | SQLite

Frameworks and Libraries: OpenCV | ROS | React | Node-RED | FastAPI | Streamlit | PyQt | TensorFlow | PyTorch

Tools: Git | GitHub | Docker | Xilinx Vivado

Technical Fields: Embedded Systems | Robotics | IoT | Machine Learning | Computer Vision | DevOps

Hardware Platforms: Arduino | ESP32 | Raspberry Pi | STM32 | Nvidia Jetson Orin

JetVIO*Python | C++ | ROS*

July 2025 – June 2026

Final Year Project (Group)

- An autonomous UAV that uses visual-inertial navigation for localization in GPS-denied environments.
- Interfaced Intel RealSense and Orbbec Gemini 2L cameras with ROS2 to extract visual and depth information for state estimation.
- Studied deep learning-based image enhancement techniques to improve visual odometry performance in low-light conditions.

FPGA Audio Spectrum Analyzer*VHDL | Xilinx Vivado | Nexys A7 FPGA*

Jul 2025 – Dec 2025

Module Project (Group)

- Real-time hardware-accelerated audio visualizer performing FFT analysis and VGA rendering on an FPGA.
- Experimented with audio input subsystem using UART and I2S protocols.
- Deployed on Nexys A7 FPGA Board and achieved low-latency signal processing and synchronized playback.

QUOT3D – Automating 3D Printing Quotation*Python | FastAPI | Docker | Git*

Jan 2025 – Jun 2025

Module Project (Group)

- Web-based platform for generating instant and accurate 3D printing quotations from uploaded 3D models by slicing the model.
- Developed the backend service to analyze 3D model files and extract pricing-related metrics.
- Enabled automated, vendor-configurable quotation generation, reducing manual communication overhead.

FazeTrak – Smart Webcam*Python | C++ | OpenCV | PyQt | ESP32*

Jul 2024 – Nov 2024

Semester Project (Individual)

- Smart webcam system with face recognition and gesture-based face locking and tracking.
- Implemented computer vision pipeline, servo control logic, and desktop application for hardware integration.
- Achieved real-time face tracking with pan-tilt control and streaming video for usage of other applications via OBS Virtual Camera Driver.

MinerShield*C++ | JavaScript | ESP32 | Node-RED*

Jul 2024 – Oct 2024

Competition Project (Group)

- IoT-based wireless monitoring system for miner safety in underground environments.
- Contributed to embedded system integration, IoT workflow development and frontend development.
- Demonstrated reliable environmental and miner-status monitoring through a prototype sensor network.

MechaPulse*C++ | ESP32 | Raspberry Pi | Python | FastAPI*

Jan 2024 – May 2024

Competition Project (Group)

- Industrial IoT system for machinery fault detection using acoustic signal analysis via Machine Learning.
- Led the deployment of ML models on Raspberry Pi and managed embedded system integration.
- Validated fault detection of a drill machine using the prototype system.

Smart MediBox*Arduino | ESP32 | IoT | PCB Design*

Jan 2024 – Mar 2024

Module Project (Individual)

- Smart medicine container for indicating scheduled medicine intake times.
- Designed embedded firmware, IoT connectivity, and Node-RED based control dashboard.
- Delivered a functional prototype supporting remote monitoring and timely medication alerts.

AspireTrust – Bank Transaction and Loan Processing System*MySQL | HTML | CSS | Git*

Sep 2023 – Nov 2023

Module Project (Group)

- Database-driven banking system supporting transactions and loan processing.
- Contributed to database schema design and query implementation.
- Produced a simple prototype system demonstrating robust database management practices.

Nanoprocessor*VHDL | Xilinx Vivado | Basys3 FPGA*

May 2023 – Jun 2023

Module Project (Group)

- Custom-designed 4-bit nanoprocessor implemented on FPGA hardware.
- Architected the top-level processor design and managed simulation workflows.
- Successfully deployed and validated the processor on a Basys3 FPGA board.

CERTIFICATIONS

Embedded AI	Getting Started with Machine Learning at the Edge: <i>Arm</i> <i>Apr 2025</i> ↗
	Computer Vision with Embedded Machine Learning: <i>Edge Impulse</i> <i>Mar 2025</i> ↗
	Introduction to Embedded Machine Learning: <i>Edge Impulse</i> <i>Mar 2025</i> ↗
Computer Vision & AI	Vision Language Models Bootcamp: <i>OpenCV University</i> <i>Dec 2025</i> ↗
	Fundamentals of Deep Learning: <i>NVIDIA DLI</i> <i>Dec 2025</i> ↗
	Computer Vision: <i>Kaggle</i> <i>Dec 2024</i> ↗
	Intro to Machine Learning: <i>Kaggle</i> <i>Jan 2024</i> ↗
Embedded Systems	Arm Cortex-M Architecture & Software Development: <i>Arm</i> <i>Apr 2025</i> ↗
	Embedded Software & Hardware Architecture: <i>CU Boulder</i> <i>Mar 2025</i> ↗
DevOps & Engineering	GitHub Foundations: <i>GitHub</i> <i>Mar 2025</i> ↗
	Docker Training for Absolute Beginners: <i>KodeKloud</i> <i>Feb 2025</i> ↗

EXTRA CURRICULAR ACTIVITIES

IEEE Robotics and Automation Society Student Branch, UoM

• Term 24/25 - Secretary	Oct 2024 – Oct 2025
• Term 23/24 - Vice Chairman	Oct 2023 – Oct 2024
• Term 22/23 - Member—Design Committee	Oct 2022 – Oct 2023
• BotTalks 1.0 - Co Chair	Mar 2023 – Apr 2023

IEEE Student Branch, University of Moratuwa

• Term 23/24 - Member—Membership Development Committee	Nov 2023 – Nov 2024
• IEEE Open Week - Volunteer—Chapter Coordination Committee	Feb 2024 – Mar 2024
• MoraForesight 1.0 - Volunteer—Finance Committee, Design Committee	Mar 2023 – Aug 2023

Mora Hiking Club, University of Moratuwa

• Stridian 2025 - Media Team	July 2025 – Sep 2025
• Stridian 2024 - Health Team	July 2024 – Sep 2024
• Stridian 2023 - Participant	July 2023 – Sep 2023

Department of Computer Science and Engineering, University of Moratuwa

• CSE40 - Volunteer—Marketing Committee	Jan 2025 – May 2025
• Hit The Grounds - Volunteer—Marketing Committee	Dec 2024 – Jan 2025
• CSE Career Fair 2024 - Volunteer—Company Coordination Committee	Dec 2023 – Jan 2024

Techno 2023 Exhibition, IESL IT and Computer Engineering

• Graphic Designer	Sep 2023 – Oct 2023
• Demonstrator	Sep 2023 – Oct 2023

Exmo 2023 Exhibition, University of Moratuwa

• Event Committee Member	Mar 2023 – Jul 2023
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AWARDS & ACHIEVEMENTS

Dean's List: Semester 6

Evolve IoT Mini Research Competition: 1st Place

SLIoT Challenge 2023: Semifinalists

President's Scout Award: Issued in 2019

National Junior Science Olympiad 2016: Gold Medal

International Mathematics and Science Olympiad 2013: Silver Medal

National Mathematics and Science Olympiad 2013: Bronze Medal

REFERENCES

Prof. Dulani Meedeniya

Department of CSE

University of Moratuwa.

Email: dulanim@cse.mrt.ac.lk

Dr. Sulochana Sooriyaarachchi

Department of CSE

University of Moratuwa.

Email: sulochanas@cse.mrt.ac.lk